

Exercise 2

Convex Functions (1 pt)

Simple functions (0.5 pt)

Are the following functions convex? Prove your statement.

1. $f(x) = x^2, x \in \mathbb{R}$
2. $f(x) = e^{x^2}, x \in \mathbb{R}$
3. $f(x, y) = x^2 + 3xy + 2y^2, x \in \mathbb{R}, y \in \mathbb{R}$

1.

$$\begin{aligned} f(x) &= x^2 \\ \implies f'(x) &= 2x \\ \implies f''(x) &= 2 > 0 \quad \forall x \in \mathbb{R} \\ \therefore f &\text{ is convex} \end{aligned}$$

2.

$$\begin{aligned} f(x) &= e^{x^2} \\ \implies f'(x) &= 2xe^{x^2} \\ \implies f''(x) &= 2e^{x^2} + (2x)e^{x^2}(2x) = 2e^{x^2}(2x^2 + 1) > 0 \quad \forall x \in \mathbb{R} \\ \therefore f &\text{ is convex} \end{aligned}$$

3.

$$\begin{aligned} f(x, y) &= x^2 + 3xy + 2y^2 \\ \implies \nabla f(x, y) &= (2x + 3y, 3x + 4y) \\ \implies \nabla^2 f(x, y) &= \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \end{aligned}$$

$\nabla^2 f(x, y)$ is not positive semi-definite since its eigenvalues are $3 + \sqrt{10}$ and $3 - \sqrt{10} < 0$. Therefore f is not convex.

Log-sum-exp (0.5 pt)

Show that the function

$$f(\mathbf{x}) = \log(e^{x_1} + \dots + e^{x_n})$$

is convex on $\mathbb{R}^n$.

Hint: The proof is outlined in Boyd's book p. 74. Develop it yourself and provide the intermediate steps.

Geometric Mean (Bonus (0.5 pt))

Show that the geometric mean

$$f(\mathbf{x}) = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

is concave on $\mathbb{R}_{++}^n$.

Solution: One possible way is to analyze the Hessian (Boyd's book p. 74) but there is a more elegant solution.

We will use the AM-GM inequality: The arithmetic mean is greater or equal than the geometric mean.

$$\frac{1}{n} \sum_{i=1}^n a_i \geq \left(\prod_{i=1}^n a_i \right)^{1/n}$$

Replace in the previous equation the following two values:

$$a_i = \frac{x_i}{\lambda x_i + (1-\lambda)y_i} \quad b_i = \frac{y_i}{\lambda x_i + (1-\lambda)y_i}$$

To obtain two inequalities:

$$\left(\prod_{i=1}^n \frac{x_i}{\lambda x_i + (1-\lambda)y_i} \right)^{1/n} = \frac{f(\mathbf{x})}{f(\lambda \mathbf{x} + (1-\lambda)\mathbf{y})} \leq \frac{1}{n} \sum_{i=1}^n \frac{x_i}{\lambda x_i + (1-\lambda)y_i} \quad (1)$$

$$\left(\prod_{i=1}^n \frac{y_i}{\lambda x_i + (1-\lambda)y_i} \right)^{1/n} = \frac{f(\mathbf{y})}{f(\lambda \mathbf{x} + (1-\lambda)\mathbf{y})} \leq \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\lambda x_i + (1-\lambda)y_i} \quad (2)$$

Now multiply (1) by λ and (2) by $(1-\lambda)$:

$$\frac{\lambda f(\mathbf{x})}{f(\lambda \mathbf{x} + (1-\lambda)\mathbf{y})} \leq \frac{1}{n} \sum_{i=1}^n \frac{\lambda x_i}{\lambda x_i + (1-\lambda)y_i} \quad (3)$$

$$\frac{(1-\lambda)f(\mathbf{y})}{f(\lambda \mathbf{x} + (1-\lambda)\mathbf{y})} \leq \frac{1}{n} \sum_{i=1}^n \frac{(1-\lambda)y_i}{\lambda x_i + (1-\lambda)y_i} \quad (4)$$

Now sum both (3) and (4):

$$\begin{aligned}\frac{\lambda f(x_i) + (1 - \lambda)f(y_i)}{f(\lambda x_i + (1 - \lambda)y_i)} &\leq \frac{1}{n} \sum_{i=1}^n \left(\frac{\lambda x_i}{\lambda x_i + (1 - \lambda)y_i} + \frac{(1 - \lambda)y_i}{\lambda x_i + (1 - \lambda)y_i} \right) = 1 \\ \frac{\lambda f(x_i) + (1 - \lambda)f(y_i)}{f(\lambda x_i + (1 - \lambda)y_i)} &\leq 1 \\ \implies \lambda f(x_i) + (1 - \lambda)f(y_i) &\leq f(\lambda x_i + (1 - \lambda)y_i)\end{aligned}$$

Now take $g(\mathbf{x}) = -f(\mathbf{x})$ and replace:

$$g(\lambda x_i + (1 - \lambda)y_i) \leq \lambda g(x_i) + (1 - \lambda)g(y_i)$$

Which is the definition of a convex function. Since $g(\mathbf{x}) = -f(\mathbf{x})$, then $f(\mathbf{x})$ is a concave function.